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Discrete Mathematics

PART 1

COMBINATORIAL THEORY

(Lý thuyết tổ hợp)

PART 2

GRAPH THEORY

(Lý thuyết đồ thị)

Contents of Part 1

Chapter 0: Sets, Relations

Chapter 1: Counting problem

Chapter 2: Existence problem

Chapter 3: Enumeration problem

Chapter 4: Combinatorial optimization problem

Contents

- 1. Basic counting principles**
2. Elementary combinatorial configuration
3. The inclusion-exclusion principle
4. Recurrence relation
5. Generating function

1. Basic counting principles

1.1. The sum rule

1.2. The product rule

1.1. The sum rule

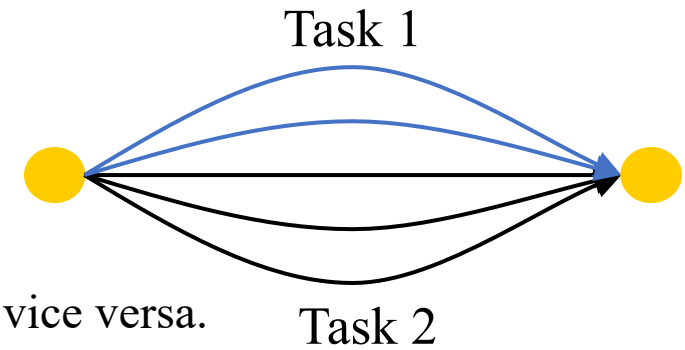
Let us consider two tasks:

m_1 is the number of ways to do **task 1**

m_2 is the number of ways to do **task 2**

Tasks are independent of each other, i.e.,

Performing **task 1** does not accomplish **task 2** and vice versa.



Sum rule: the number of ways that “**either** task 1 **or** task 2 can be done, but **not both**”, is $m_1 + m_2$.

Generalizes to multiple tasks ...

Things	1	2	3	...	k
ways	m_1	m_2	m_3	...	m_k

select one of them: $m_1 + m_2 + m_3 + \dots + m_k$ ways

1.1. The sum rule

Generalized sum rule: If we have tasks $T_1, T_2, \dots, T_k$ that can be done in $m_1, m_2, \dots, m_k$ ways, respectively, and any two of these tasks can not be done at the same time, then there are $m_1 + m_2 + \dots + m_k$ ways to do one of these tasks.

➔ The sum rule can also be phrased in terms of *set theory*: The size of the union on k finite pair wise disjoint sets is the sum of their sizes:

- Let $A_1, A_2, \dots, A_k$ be disjoint sets. Then the number of ways to choose any element from one of these sets is

$$\underline{|A_1 \cup A_2 \cup \dots \cup A_k|} = \underline{|A_1| + |A_2| + \dots + |A_k|}.$$

1.1. The sum rule

Example 1: A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects respectively. How many possible projects are there to choose from?

$$23+15+19 \quad \text{☺}$$

Example 2: How many strings of 4 decimal digits, have exactly three digits that are 9s?

The string can have:

- The non-9 as the first digit (**x999**)
- OR the non-9 as the second digit (**9x99**)
- OR the non-9 as the third digit (**99x9**)
- OR the non-9 as the fourth digit (**999x**)

Thus, we use the sum rule:

- For each of those cases, there are 9 possibilities for the non-9 digit (any number other than 9)
- Thus, the answer is $9+9+9+9 = 36$

1. Basic counting principles

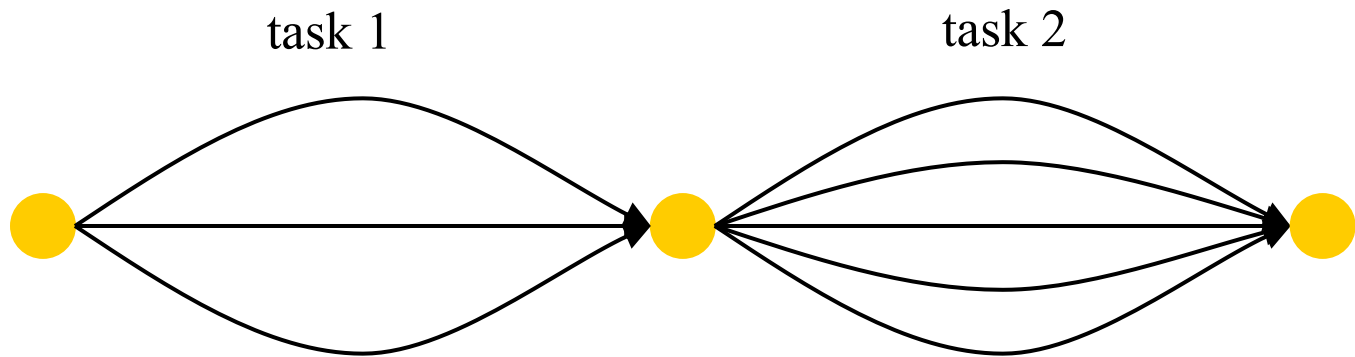
1.1. The sum rule

1.2. The product rule

The product rule

- Consider two tasks:
 - m_1 is the number of ways to do task 1
 - m_2 is the number of ways to do task 2
 - Tasks are independent of each other, i.e.,
 - Performing task 1 does not accomplish task 2 and vice versa.

Product rule: the number of ways that “both tasks 1 and 2 can be done” is $m_1 m_2$.



- Generalize to multiple tasks ...

1.2. The product rule

Generalized product rule: If we have a procedure consisting of sequential tasks $T_1, T_2, \dots, T_k$ that can be done in $m_1, m_2, \dots, m_k$ ways, respectively, then there are $m_1 * m_2 * \dots * m_k$ ways to carry out the procedure.

→ The product rule can also be phrased in terms of *set theory*: Let $A_1, A_2, \dots, A_k$ be finite sets. Then the number of ways to choose one element from each set in the order of $A_1, A_2, \dots, A_k$ is

$$|A_1 \times A_2 \times \dots \times A_k| = |A_1| * |A_2| * \dots * |A_k|.$$

1.2. The product rule

- If each element a_i of k -tuple $(a_1, a_2, \dots, a_k)$ has m_i number of ways to select ($i = 1, 2, \dots, k$), then the number of tuples could be generated is the product of these $m_1 m_2 \dots m_k$

1.2. The product rule

In many counting problems, it is only after building the first element, we just know how to build the second element; after building the first two elements we only know how to build the third element,... In that case, we use the general product rules:

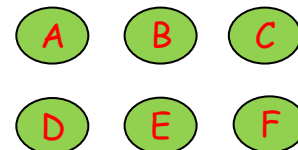
Suppose we construct a k -tuple $(a_1, a_2, \dots, a_k)$ by building each element in turn and

- a_1 can be chosen by m_1 way;
- After a_1 is selected, a_2 can be chosen by m_2 way;
- ...
- After $a_1, a_2, \dots, a_{k-1}$ are chosen, a_k can be chosen by m_k ways;

Then the number of k -tuples generated is $m_1 m_2 \dots m_k$

Example 1: 6 persons are competing for 4 prizes. How many different outcomes are possible?

1st prize	2nd prize	3rd prize	4th prize
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A 4-tuple = (1st prize, 2nd prize, 3rd prize, 4th prize)

1.2. The product rule

Example 2: There are 18 math majors and 325 CS majors. How many ways are there to pick one math major and one CS major?

$$\text{Total is } 18 * 325 = 5850$$

Example 3: The license plate: 2 letters-4 digits

(a) no letter or digit can be repeated

$$26 \times 25 \times 10 \times 9 \times 8 \times 7$$

(b) with repetitions allowed

$$26 \times 26 \times 10 \times 10 \times 10 \times 10$$

(c) same as (b), but only vowels and even digits

$$5^2 \times 5^4$$

More complex counting problems

- Combining the **product rule** and the **sum rule**.
- Thus we can solve more interesting and complex problems.

More complex counting problems

Example 1: How many BASIC variables: single letter

or single letter+single digit

$$26 + 26 * 10 = 286$$

rule of sum rule of product

More complex counting problems

Example 2: How many integers from 1 through 999 do not have any repeated digits?

Let A = integers from 1 to 999 not having repeated digits.

Partition A into 3 sets:

- A_1 =one-digit integers not having repeated digits;
- A_2 =two-digit integers not having repeated digits;
- A_3 =three-digit integers not having repeated digits.

$$\begin{aligned}\rightarrow |A| &= |A_1| + |A_2| + |A_3| && \text{(by the sum rule)} \\ &= 9 + 9 \times 9 + 9 \times 9 \times 8 = 738 && \text{(by the product rule)}\end{aligned}$$

Contents

1. Basic counting principles
- 2. Elementary combinatorial configuration**
3. The inclusion-exclusion principle
4. Recurrence relation
5. Generating function

2. Elementary combinatorial configuration

2.1. Permutation

2.2. Combination

2. Elementary combinatorial configuration

2.1. Permutation

2.1.1. Permutation

2.1.2. Circulation permutation

2.1.3. Permutation of multisets



2.2. Combination

2.1.1. Permutation

Example 1: A PIN is defined as a sequence of any 4 digits from the set $\{0, 1, \dots, 9\}$.

Question 1. How many different PINs are possible if *repetitions are allowed*?

Solution. Choosing a PIN is a 4-step operation:

Step 1: Choose the 1st symbol (10 different ways).

Step 2: Choose the 2nd symbol (10 different ways).

Step 3: Choose the 3rd symbol (10 different ways).

Step 4: Choose the 4th symbol (10 different ways).

➔ Based on the *product rule*:

$10 \times 10 \times 10 \times 10 = 10,000$ PINs are possible.

2.1.1. Permutation

Example 1: A PIN is defined as a sequence of any 4 digits from the set $\{0, 1, \dots, 9\}$.

Question 2. How many different PINs are possible if *repetitions are NOT allowed*?

Solution. Choosing a PIN is a 4-step operation:

Step 1: Choose the 1st symbol (10 different ways).

Step 2: Choose the 2nd symbol (9 different ways).

Step 3: Choose the 3rd symbol (8 different ways).

Step 4: Choose the 4th symbol (7 different ways).

➔ Based on the *product rule*:

$10 \times 9 \times 8 \times 7 = 5,040$ PINs are possible.

2.1.1. Permutation

A *permutation* of a set A of objects is an ordered arrangement of the elements of A where each element appears only once

Example: If $A = \{a, b, c\}$, then the permutations of A are

1. abc
2. acb
3. bac
4. bca
5. cab
6. cba

The number of permutations of any set with n elements is

$$P(n) = n! = n(n - 1) \cdot \cdot \cdot 2 \cdot 1$$

(Note that by definition $0! = 1$)

2. Elementary combinatorial configuration

2.1. Permutation

2.1.1. Permutation

2.1.2. Circulation permutation

2.1.3. Permutation of multisets

2.2. Combination



2.1.2. Circulation permutation

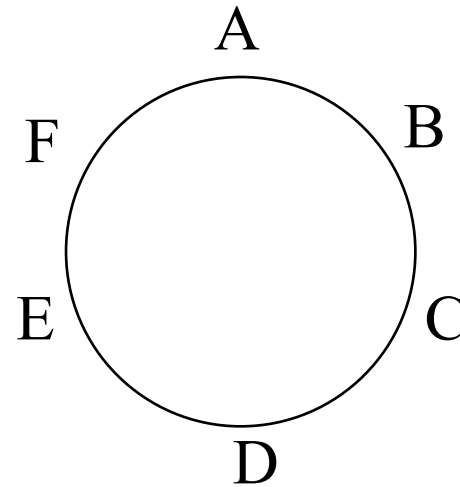
Circulation permutation $n!/n=(n-1)!$

Example 3: 6 people A, B, C, D, E, F are seated around a round table, how many different circular arrangements are possible, if arrangements are considered the same when one can be obtained from the other by rotations?

ABCDEF,
BCDEFA,
CDEFAB,
DEFABC,
EFABCD,
FABCDE

are the same arrangements circularly

➔ There are $6!$ ways to seat 6 people around the table
For each seating, there are 6 “rotations” of the seating
Thus, the final answer is $6!/6 = 5! = 120$



2. Elementary combinatorial configuration

2.1. Permutation

2.1.1. Permutation

2.1.2. Circulation permutation

2.1.3. Permutation of multisets



2.2. Combination

2.1.4. Permutations of multisets

A *multiset* M is a collection whose members need not be distinct.

Example: The collection

$$M = (a, a, a, b, b, c, d, d, d, 1, 2, 2, 2, 3, 3, 3, 3)$$

is a multiset; and sometimes it is convenient to write

$$M = (3a, 2b, c, 3d, 1, 3\bullet 2, 4\bullet 3).$$

A multiset M over a set S can be viewed as a function $v : S \rightarrow \mathbb{N}$ from S to the set $\mathbb{N}$ of nonnegative integers; each element $x \in S$ is repeated $t(x)$ times in M ; we write $M = (S; t)$.

2.1.4. Permutation of multisets

Let M be a multiset and $|M| = n$.

Proposition 1. *Let M be a multiset of r different types where each type has infinitely elements. Then the number of k -permutations of M equals r^k*

Example 1. What is the number of binary numerals with at most 4 digits?

Solution: The question is to find the number of 4-permutations of the multiset $(\infty 0, \infty 1)$. Thus the answer is $2^4 = 16$.

2.1.4. Permutation of multisets

Proposition 2. *Let M be a multiset of r different types with repetition numbers $n_1, n_2, \dots, n_r$ respectively. Let $n = n_1 + n_2 + \dots + n_r$. Then the number of permutations of M equals*

$$\frac{n!}{n_1! n_2! \dots n_r!}$$

Proof.

n_r

2.1.4. Permutation of multisets

Example: Find the number of 8-permutations of the multiset $M = (a, a, a, b, b, c, c, c) = (3a, 2b, 4c)$.

Solution:

How to select 8 elements from $(3+2+4) = 9$ elements ???

$(2a, 2b, 4c)$

$(3a, b, 4c)$

$(3a, 2b, 3c)$

The number of 8-permutations of $(2a, 2b, 4c)$: $8!/(2!2!4!)$

The number of 8-permutations of $(3a, b, 4c)$: $8!/(3!1!4!)$

The number of 8-permutations of $(3a, 2b, 3c)$: $8!/(3!2!3!)$

→ Thus the answer is

$$8!/(2!2!4!) + 8!/(3!1!4!) + 8!/(3!2!3!) = 420 + 280 + 560 = 1260$$

2. Elementary combinatorial configuration

2.1. Permutation

2.2. Combination

2.2. Combination

2.1.1. Definitions

2.2.2. Binomial coefficients

2.2.3. Combinations of Multisets

2.2.4. Multinomial coefficients

2.2.1. Definitions

A k -combination of a set of n elements is a subset of size k of n elements.

(Note: A permutation is a sequence while **a combination is a set**)

Example:

The **2-permutation** (**sequence**) of SOHN is:

SO, SH, SN, OH, ON, **OS**, HN, HS, HO, NS, NO, NH

The **2-combination** (**set**) of SOHN is:

{S,O}, {S,H}, {S,N}, {O,H}, {O,N}, {H,N}

2.2.1. Definitions

The number of all k -combinations of a set of n elements denoted

$$\binom{n}{k}, C_n^k \text{ or } C(n, k)$$

and read “ n choose k ”.

$$\binom{n}{k} = \frac{n!}{k! \cdot (n - k)!}$$

Proof ???

This number is also called a *binomial coefficient* because such numbers occur as coefficients in the expansions of powers of binomial expressions such as $(a+b)^n$

k -combination

Example: $X = \{a, b, c\}; n = 3, r = 2$

k -permutation

select 2 from 3 elements of X :

1. (a, b)
 2. (a, c)
 3. (b, a)
 4. (b, c)
 5. (c, a)
 6. (c, b)
- Ordered sequence

2!

k -combination

select 2 from 3 elements of X :

1. $\{a, b\}$
 2. $\{a, c\}$
 3. $\{b, c\}$
- Order is not the matter

$$P_n^k = n(n-1)\dots(n-k+1) = \frac{n!}{(n-k)!}$$

$$C_n^k = \frac{n!}{k!(n-k)!}$$

Difference between permutation and combination

Example: A club has 25 members.

- a. How many ways are there to choose 4 members of the club to serve on an executive committee?

These 4 members are unordered, so use combinations:

$$C(25,4) = 25!/4!(25-4)! = 25*23*22 = 12,650$$

- b. How many ways are there to choose a president, vice president, secretary, and treasurer of the club?

These selections are ordered, so use permutations:

$$P(25,4) = 25!/(25-4)! = 25*24*23*22 = 303,600$$

2.2. Combination

2.1.1. Definitions

2.2.2. Binomial coefficients

2.2.3. Combinations of Multisets

2.2.4. Multinomial coefficients

Binomial Coefficients

$$(a + b)^4 = (a + b)(a + b)(a + b)(a + b)$$

$$= \binom{4}{0} a^4 + \binom{4}{1} a^3 b + \binom{4}{2} a^2 b^2 + \binom{4}{3} a b^3 + \binom{4}{4} b^4$$

Binomial Theorem: Let x and y be variables, and let n be any nonnegative integer. Then

$$(x + y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j$$

2.2. Combination

2.1.1. Definitions

2.2.2. Binomial coefficients

2.2.3. Combinations of Multisets

2.2.4. Multinomial coefficients

2.2.3. Combinations of multisets

Let M be a multiset $\{\infty a_1, \infty a_2, \dots, \infty a_n\}$ (M has n distinct objects):

A k -combination of M is an unordered collection of k objects selected from n types of objects of M .

A k -combination of M is also called an k -combination with repetition allowed.

The number of k -combination of M is $C(n+k-1, k) = C(n+k-1, n-1)$

(the number of selections, **with repetitions**, of k objects from n distinct objects)



Need to divide k candies for n kids $B_1, B_2, \dots, B_n$. How many different ways to divide?

Let t_j be the number of candies for kid $B_j, j=1, \dots, n$. At this point, the above problem leads to the problem:

Let k and n be non-negative integers. How many non-negative integers in the following equation have?

$$t_1 + t_2 + t_3 + \dots + t_n = k$$

$$t_1, t_2, \dots, t_n \in \mathbb{Z}^+$$

The number of r -combination of M is $C(n+ r-1, r)$

Proof: When k objects are selected from the multiset M , we put them into the following n boxes

1st	2nd	...	...	nth

so that the i th type objects are contained in the i th box, $1 \leq i \leq n$.

Since the objects of the same type are identical, we may use the symbol "O" to denote any object in the boxes, and the objects in different boxes are separated by a stick "|".

For example, for $n = 4$ and $k = 7$:

$\{a, a, b, c, c, c, d\}$

$\longleftrightarrow$

OO	O	OOO	O
1	2	3	4

$\longleftrightarrow$

0010100010

$\{b, b, b, b, d, d, d\}$

$\longleftrightarrow$

	OOOO		OOO
1	2	3	4

$\longleftrightarrow$

1000011000

Convert the symbol "O" to zero 0 and the stick "|" to one 1, any such placement is converted into a 0-1 sequence of length $k+n-1$ with exactly k zeros and $n-1$ ones

➔Now the problem becomes counting the number of 0-1 words of length $k+(n-1)$ with exactly k zeros and $n-1$ ones

$$= C(n+ k-1, k) = C(n+ k-1, n-1)$$

2.3.3. Combinations of multisets

Example 1: How many ways to divide 10 candies for 4 kids
(there may be kids without any candies)

How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 = 10$$

when the variables are nonnegative integers ($x_i \geq 0$)?

Answer:

Select with repetition from x_1, x_2, x_3, x_4 10 times.

For example if x_1 is selected twice then $x_1 = 2$ in the final solution.

Therefore: **k -combination with repetition allowed** ($n = 4, k = 10$):

$C(4+10-1, 10)$

2.3.3. Combinations of multisets

Example 2: How many nonnegative integer solutions are there to the inequality

$$x_1 + x_2 + \cdots + x_6 < 10?$$

It is equivalent to

$$x_1 + x_2 + \cdots + x_6 + x_7 = 10, \quad 0 \leq x_i, \quad 1 \leq i \leq 6, \quad 0 < x_7$$

which can be transformed to

$$y_1 + y_2 + \cdots + y_6 + y_7 = 9, \quad y_i \geq 0 \text{ integer}$$

where $y_i = x_i$ for $1 \leq i \leq 6$

and $y_7 = x_7 - 1$

➔ Answer: 7 types of objects to select $(y_1, \dots, y_7)$ and 9 times of selections $[k = 9, n = 7]$

➔ k -combination with repetition $C(7+9-1, 9) = C(15, 9) = 5005$

2.2. Combination

2.1.1. Definitions

2.2.2. Binomial coefficients

2.2.3. Combinations of Multisets

2.2.4. Multinomial coefficients

2.2.4. Multinomial coefficients

The number of ordered arrangements of n objects, in which there are k_1 objects of type 1, k_2 objects of type 2, ..., and k_m objects of type m and where $k_1 + k_2 + \dots + k_m = n$, is

$$\binom{n}{k_1, k_2, \dots, k_m} = \frac{n!}{k_1! k_2! \dots k_m!}$$

2.2.4. Multinomial coefficients

The Multinomial Theorem:

$$(x_1 + x_2 + \dots x_m)^n = \sum \binom{n}{k_1, k_2, \dots, k_m} x_1^{k_1} x_2^{k_2} \dots x_m^{k_m}$$

where the summation is over all sequences of non-negative integers $(k_1, k_2, \dots, k_m)$ such that $k_1 + k_2 + \dots + k_m = n$.

Summary:

	Order matters	Order does not matter
Repetition not allowed	1 $P(n, k) = \frac{n!}{(n - k)!}$ (<i>k</i>-permutation of <i>n</i>-set)	2 $C(n, k) = \frac{n!}{k! (n - k)!}$ (<i>k</i>-combination of <i>n</i>-set)
Repetition allowed	3 n^k (<i>k</i>-permutation with repetition allowed)	4 $C(n + k - 1, k) = \frac{(n + k - 1)!}{k! (n - 1)!}$ (<i>k</i>-combination with repetition allowed)

The set A = {S,O,H,N}	
The 2-permutation (sequence) of set A is: P(4,2) = 12 SO, SH, SN, OH, ON, OS, HN, HS, HO, NS, NO, NH 1	The 2-combination (set) of set A is: C(4, 2) = 6 {S,O},{S,H},{S,N},{O,H},{O,N},{H,N} 2
The multi set B = {∞S, ∞O, ∞H, ∞N}	
The 2-permutation with repetition allowed of set B is: 4² = 16 SS, OO, HH, NN, SO, SH, SN, OH, ON, OS, HN, HS, HO, NS, NO, NH 3	The 2-combination with repetition allowed of set B is: C(4+2-1,2)=10 {S, S}, {O, O}, {H, H}, {N, N}, {S,O},{S,H},{S,N},{O,H},{O,N},{H,N} 4

Contents

1. Basic counting principles
2. Elementary combinatorial configuration
- 3. The inclusion-exclusion principle**
4. Recurrence relation
5. Generating function

3. Inclusion-exclusion principle

3.1. Inclusion-exclusion principle

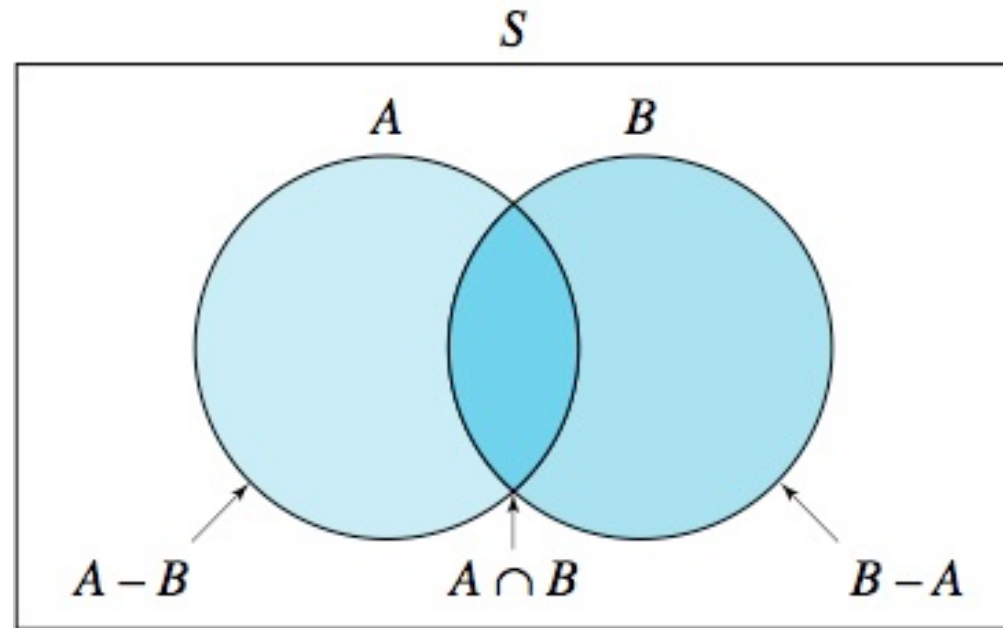
3.2. Derangement

3.1. Inclusion-exclusion principle

The inclusion–exclusion principle is an equation relating the sizes of two sets and their union. It states that if A and B are two (finite) sets, then

$$|A \cup B| = |A| + |B| - |A \cap B|$$

The meaning of the statement is that the number of elements in the union of the two sets is the sum of the elements in each set, respectively, minus the number of elements that are in both.



Similarly, for three sets A , B and C :

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

3.1. Inclusion-exclusion principle

More generally, for finite sets $A_1, A_2, \dots, A_m$

$$|A_1 \cup A_2 \cup \dots \cup A_m| = N_1 - N_2 + \dots + (-1)^{m+1} N_m$$

where:

$$N_k = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq m} |A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}|, \quad k = 1, 2, \dots, m$$

Note: N_k is the sum of the cardinalities of all intersections of k from m given sets. For example:

$$N_1 = |A_1| + \dots + |A_m|$$

$$N_m = |A_1 \cap A_2 \cap \dots \cap A_m|$$

3.1. Inclusion-exclusion principle

Example: How many integers from 1 to 1000 are either multiples of 3 or multiples of 5?

A = set of all integers from 1 to 1000 that are multiples of 3.

B = set of all integers from 1 to 1000 that are multiples of 5.

$A \cup B = ?$

The set of all integers from 1 to 1000 that are multiples of either 3 or 5.

$A \cap B = ?$

The set of all integers that are both multiples of 3 and 5, which also is the set of integers that are multiples of 15.

To use the inclusion-exclusion principle to obtain $|A \cup B|$, we need

$|A|$

$|B|$

$|A \cap B|$

3.1. Inclusion-exclusion principle

A = set of all integers from 1 to 1000 that are multiples of 3 $\rightarrow |A| = ?$

From 1 to 1000, every third integer is a multiple of 3, each multiple of 3 is of the form $3p$, for any integer p from 1 through $[1000/3]=333$, Hence $|A| = 333$.

B = set of all integers from 1 to 1000 that are multiples of 5 $\rightarrow |B| = ?$

Similarly for multiples of 5, each multiple of 5 is of the form $5q$ for some integer q from 1 through $[1000/5]=200$. Hence, we have $|B| = 200$.

$A \cap B$: set of all integers from 1 to 1000 that are multiples of 15 $\rightarrow |A \cap B| = ?$

To determine the number of multiples of 15 from 1 through 1000, each multiple of 15 is of the form $15r$ for some integer r from 1 through $[1000/15]=66$. $\rightarrow |A \cap B| = 66$.

From the principle, we have the number of integers either multiples of 3 or multiples of 5 from 1 to 1000 given by

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= 333 + 200 - 66 = 467 \end{aligned}$$

3. Inclusion-exclusion principle

3.1. Inclusion-exclusion principle

3.2. Derangement

3.2. Derangement

A **derangement** of $\{1, 2, \dots, n\}$ is a permutation on the set such that none of elements i is placed at position i th in the permutation.

(In other words, derangement is a permutation that has no fixed points)

Example: $n = 5$

Permutation $(2, 3, 5, 4, 1)$ is not a derangement

Permutation $(2, 3, 5, 1, 4)$ is a derangement

Denote D_n the number of derangements of $\{1, 2, \dots, n\} \rightarrow D_n = ??$

3.2. Derangement

Denote D_n the number of derangements of $\{1, 2, \dots, n\} \rightarrow D_n = ??$

Let S = set of all permutations of $\{1, 2, \dots, n\} \rightarrow |S| = n!$

Let A_i = subset of permutations of $\{1, 2, \dots, n\}$ such that the i th element = i in the permutation.
 $|A_1 \cup A_2 \cup \dots \cup A_n| ???$

$\rightarrow |A_1 \cup A_2 \cup \dots \cup A_n|$ counts the number of permutations in which at least one object i of the n objects appears in the i th position (**its original position**).

$\rightarrow D_n$: number of permutations such that none of the n objects appears in their original positions. Therefore:

$$D_n = n! - |A_1 \cup A_2 \cup \dots \cup A_n|$$

Remember the inclusion-exclusion principle:

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| = & \sum_{i=1}^n |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| \\ & - \sum_{i < j < k < \ell} |A_i \cap A_j \cap A_k \cap A_\ell| + \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n| \end{aligned}$$

3.2. Derangement

An example of derangements arises in a very famous problem called the **hat-check problem**:

*“ n people walk into a party and give their hats to **hat-check** girl. Unfortunately, she completely loses track of which of n hats belong to which owners. Therefore, when the party finishes, their hats are returned at random.”*

Question: What is the probability that nobody receives their own hat back?

→ This is equivalent to asking **for the probability that a permutation of n objects is a derangement.**

$$\rightarrow = \frac{D_n}{n!}$$

What is the value in number????

since there are D_n possible derangements and $n!$ possible permutations.

3.2. Derangement

It is amusing to note that as $n \rightarrow \infty$, the probability that a permutation of n values is a derangement is given by

$$\lim_{n \rightarrow \infty} P(\text{derangement}) = \lim_{n \rightarrow \infty} \frac{D_n}{n!} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} = \frac{1}{e} \simeq 0.367879.$$

➔ if n is large (in practical applications, any n greater than about 10 can be considered to be large), then the probability that the permutation of the n values is a derangement is approximately $1/e$ almost independently of the precise value of n .

n	n!	Probability	#Derangements
0	1	1	
1	1	0	0
2	2	0.5	1
3	6	0.333333333	2
4	24	0.375	9
5	120	0.366666667	44
6	720	0.368055556	265
7	5040	0.367857143	1854
8	40320	0.367881944	14833
9	362880	0.367879189	133496
10	3628800	0.367879464	1334961
Note: $1/e \simeq 0.367879441$			

Contents

1. Basic counting principles
2. Elementary combinatorial configuration
3. The inclusion-exclusion principle
- 4. Recurrence relation**
5. Generating function

4. Recurrence relations

4.1. Recurrence relations

4.2. Solve recurrence relations

4.1. Recurrence relations

Definition:

A recurrence relation for the sequence $\{a_n\}$ is the equation that expresses a_n in terms of one or more of the previous terms in the sequence namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.

A sequence is called a **solution** to a recurrence relation if its terms satisfy the recurrence relation.

Example: Consider the recurrence relation

$$a_n = 2a_{n-1} - a_{n-2} \text{ for } n = 2, 3, 4, \dots$$

The sequence $a_n = 3n$ is a solution of this recurrence relation? The sequence $a_n = n+1$??

For $n \geq 2$ we see that: $2a_{n-1} - a_{n-2} = 2(3(n-1)) - 3(n-2) = 3n = a_n$

Therefore, the sequence $a_n = 3n$ is a solution of the recurrence relation

The sequence $a_n = 5$ is also a solution of the this recurrence relation?

For $n \geq 2$ we see that: $2a_{n-1} - a_{n-2} = 2 \cdot 5 - 5 = 5 = a_n$

Therefore, the sequence $a_n = 5$ is also a solution of the recurrence relation

the same recurrence relation can have **multiple solutions**. → Why???

4.1. Recurrence relations

A recurrence **without specifying any initial values (initial conditions)**.

→ can have (and usually has) **multiple solutions**.

Example: $a_n = 2a_{n-1} - a_{n-2}$ for $n = 2, 3, 4$. It has the following sequences a_n as solution:

$$a_n = 5$$

$$a_n = 3n$$

$$a_n = n + 1$$

If **both** the initial conditions and the recurrence relation are specified, then the sequence is **uniquely** determined.

Example: $a_n = 2a_{n-1} - a_{n-2}$ for $n = 2, 3, 4$

where $a_0 = 0; a_1 = 3$

→ The sequence $a_n = 5$ is not the solution

→ The sequence $a_n = 3n$ is the unique solution

4.1. Recurrence relations

Some applications of recurrence relations:

1. To solve many counting problems we can try construct recurrence relations.
2. Complexity analysis of the recursive algorithm is a recurrence relation on the number of operations.

Modeling with Recurrence Relations

Example 1: Someone deposits \$10,000 in a savings account at a bank yielding 5% per year with interest compounded annually. How much money will be in the account after 30 years?

Solution:

- Let P_n denote the amount in the account after n years.

How can we determine P_n on the basis of P_{n-1} ?

- We can derive the following **recurrence relation**:

$$P_n = P_{n-1} + 0.05P_{n-1} = 1.05P_{n-1}.$$

- The initial condition is $P_0 = 10,000$.

$$P_{30} = (1.05)^{30} \cdot 10,000 = 43,219.42$$

Then we have:

- $P_1 = 1.05P_0$
- $P_2 = 1.05P_1 = (1.05)^2P_0$
- $P_3 = 1.05P_2 = (1.05)^3P_0$
- ...
- $P_n = 1.05P_{n-1} = (1.05)^nP_0$

➔ We now have a **formula** to calculate P_n for any natural number n and can avoid the iteration.

Example 2: Fibonacci

A pair of rabbits is placed on an island. After 2 months old, a pair of rabbits produces another pair each month. How many rabbits are there after n months?

- $n = 1: f_1 = 1$
- $n = 2: f_2 = 1$ **Fibonacci sequence: 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, etc.**
- $n > 2: f_n = f_{n-1} + f_{n-2}$

The $n > 2$ formula is true since each new pair comes from a pair at least 2 months old.

New-born rabbits	Month	1-month rabbits	>=2-month rabbits	Total f_n
	1	 1	0	1
	2	0	1	1
	3	1	1	2
 	4	1	2	3
	5	2	3	5
	6	3	5	8
	7	5	8	13
	8	8	13	21

4.1. Recurrence relations

Some applications of recurrence relations:

1. To solve many counting problems we can try construct recurrence relations.
2. Complexity analysis of the recursive algorithm is a recurrence relation on the number of operations.

4.1. Recurrence relations

Example 3. Recall the factorial function

$$n! = \begin{cases} 1 & \text{if } n = 1 \\ n \cdot (n - 1)! & \text{if } n > 1 \end{cases}$$

Consider the following (recursive) algorithm for computing $n!$

Algorithm (FACTORIAL)

```
INPUT      :  $n \in \mathbb{N}$ 
OUTPUT     :  $n!$ 
1 IF  $n = 1$  THEN
2   return 1
3 END
4 ELSE
5   return FACTORIAL( $n - 1$ )  $\times$   $n$ 
6 END
```

4.1. Recurrence relations

Factorial(n);

Algorithm (FACTORIAL)

```
INPUT      :  $n \in \mathbb{N}$ 
OUTPUT     :  $n!$ 
1 IF  $n = 1$  THEN
2   return 1
3 END
4 ELSE
5   return FACTORIAL( $n - 1$ )  $\times$   $n$ 
6 END
```

How many times is function Factorial called when we execute the statement **Factorial(n)**; ?

When $n = 1$, Factorial is called 1 $\rightarrow T(1) = 1$

Otherwise:

We perform 1.

Plus the number of calls in the recursive call in Factorial($n - 1$) $\rightarrow T(n-1)$

This can be expressed as a formula (similar to the definition of $n!$):

$$T(1) = 1$$

$$T(n) = 1 + T(n - 1), n > 1$$

$$\left. \begin{array}{l} T(1) = 1 \\ T(n) = 1 + T(n - 1), n > 1 \end{array} \right\} T(n) = 1 + T(n-1)$$

This is known as a recurrence relation.

Find the **formula** to calculate $T(n)$ for any natural number n and can avoid the iteration.

explicit formula

4.1. Recurrence relations

Factorial(n);

Algorithm (FACTORIAL)

```
INPUT      :  $n \in \mathbb{N}$ 
OUTPUT     :  $n!$ 
1 IF  $n = 1$  THEN
2   return 1
3 END
4 ELSE
5   return FACTORIAL( $n - 1$ )  $\times$   $n$ 
6 END
```

$$\begin{aligned} T(n) &= T(n-1) + 1, n > 1 \\ &= [T(n-2) + 1] + 1 \quad [\\ &= [[T(n-3) + 1] + 1] + 1 \quad [\text{has 3 ones}] \\ &= \dots \\ &= T(n - (n-1)) + (n-1) \quad [\text{has } n-1 \text{ ones}] \\ &= T(1) + (n-1) \\ &= 1 + (n-1) \\ &= n \end{aligned}$$

Backward substitution

4.1. Recurrence relations

Example 4. Performance of Recursive Binary Search

Input: An array A consists of n elements: $A[0], \dots, A[n-1]$ in ascending order; Value **key** with the same data type as array A .

Output: the index in array if **key** is found, -1 if **key** is not found

To solve this problem, the divide and conquer technique is built based on the following argument: The given **key** is either

- equal to the element at the **middle** of array A

- or equal to the element at the **left (L) half** of array A

- or equal to the element at the **right (R) half** of array A .

- or not in array

4.1. Recurrence relations

Input: An array A consists of n elements: $A[0], \dots, A[n-1]$ in ascending order;
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- equal to the element at the **middle** of array A
- or equal to the element at the **left (L) half** of array A
- or equal to the element at the **right (R) half** of array A .
- or not in array

```
int binsearch(int low, int high, int A[], int key)
{
    if (low <= high)
    {
        int mid = (low + high) / 2;
        if (A[mid]==key) return mid;
        else if (key < A[mid])
            return binsearch(low, mid-1, A, key);
        else
            return binsearch(mid+1, high, A, key);
    }
    else return -1;
}
```

4.1. Recurrence relations

Example 4. Performance of Recursive Binary Search

Input: An array A consists of n elements: $A[0], \dots, A[n-1]$ in ascending order;
Value **key** with the same data type as array A .

Output: the index in array if **key** is found, -1 if **key** is not found

```
int binsearch(int low, int high, int A[], int key)
{
    if (low <= high)
    {
        int mid = (low + high) / 2;
        if (A[mid]==key) return mid;
        else if (key < S[mid])
            return binsearch(low, mid-1, A, key);
        else
            return binsearch(mid+1, high, A, key);
    }
    else return -1;
}
```

→ $\text{binsearch}(0, n-1, A, \text{key});$

How many times is binsearch called in the worst case ?

Backward substitution

Backward substitution: this works exactly as its name suggests. Starting from the equation itself, work backwards, substituting values of the function for previous ones.

Example: $T(n) = T(n-1) + 2n$

where $T(1) = 5$

Answer: We begin by unfolding the recursion by a simple substitution of the function values.

- We observe that:

$$\begin{aligned} T(n-1) &= T((n-1) - 1) + 2(n-1) \\ &= T(n-2) + 2(n-1) \end{aligned}$$

- Substituting into the original equation:

$$T(n) = T(n-2) + 2(n-1) + 2n$$

Backward substitution

- If we continue to do that we get

$$T(n) = T(n-2) + 2(n-1) + 2n$$

$$= T(n-3) + 2(n-2) + 2(n-1) + 2n$$

$$= T(n-4) + 2(n-3) + 2(n-2) + 2(n-1) + 2n$$

.....

$$= T(n-i) + \sum_{j=0}^{i-1} 2(n-j)$$

function's value at the i^{th} iteration

- Solving the sum we get

$$T(n) = T(n-i) + 2n(i-1) - 2(i-1)(i-1+1)/2 + 2n$$

$$= T(n-i) + 2n(i-1) - i^2 + i + 2n$$

Backward substitution

- We want to get rid of the recursive term $T(n-i)$. To do that, we need to know at what iteration we reach our base case, i.e. for what value of i can we use the initial condition $T(1)=5$?

We get the base case when $n - i = 1$ or $i = n - 1$

- Substituting in the equation above we get

$$\begin{aligned} T(n) &= T(1) + 2n(n-1-1) - (n-1)^2 + (n-1) + 2n \\ &= 5 + 2n(n-2) - (n^2-2n+1) + (n-1) + 2n = n^2 + n + 3 \end{aligned}$$

4. Recurrence relations

4.1. Recurrence relations

4.2. Solving recurrence relations

4.2. Solving Recurrence Relations

Solving the recurrence relation for a sequence $a_0, a_1, a_2, a_3, \dots, a_n, \dots$ is to give the **explicit formula** to compute the value for the general term a_n , i.e., to find an expression for a_n that does not involve any other a_i

Example: Given the recurrence relation:

$$a_n = 2a_{n-1} - a_{n-2} \text{ where } n = 2, 3, 4, \dots$$

$$a_0 = 0; a_1 = 3$$

→ The explicit formula for the above recurrence relation is $a_n = 3n$

→ $a_n = 3n$ is the solution to the above recurrence relation

- Does not exist the method to solve all types of recurrence relation.
- Consider method to solve the recurrence relation with following types:
 - A linear homogeneous recurrence relation of degree k with constant coefficients
 - A linear nonhomogeneous recurrence relation of degree k with constant coefficients

4.2. Solving Recurrence Relations

Definition: A linear homogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

where $c_1, c_2, \dots, c_k$ are real number constants, and $c_k \neq 0$.

- A sequence satisfying such a recurrence relation is uniquely determined by the recurrence relation if it also satisfying k initial conditions:

$$a_0 = C_0, a_1 = C_1, a_2 = C_2, \dots, a_{k-1} = C_{k-1}$$

where $C_0, C_1, \dots, C_{k-1}$ are constants.

4.2. Solving Recurrence Relations

Definition: A linear homogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

where $c_1, c_2, \dots, c_k$ are real number constants, and $c_k \neq 0$.

Explain:

- Linear: the right-hand side is the sum of the terms before the term a_n in the sequence where the coefficients ($c_1, c_2, \dots, c_k$) are constant (not a function dependent on n)
- Homogeneous: the right-hand side has no additional terms other than the terms a_i of the sequence
- Degree k : the right hand side has the $(n-k)$ th term of the sequence

A linear homogeneous recurrence relation of degree k with constant coefficients

- We try to find solution of the form $a_n = r^n$, where r is a constant.
- The sequence $\{a_n = r^n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$$

if and only if r satisfying:

$$r^n = c_1 r^{n-1} + \dots + c_k r^{n-k}, \text{ or } \quad \text{(subtract the right-hand side from the left and } \times \text{ by } r^{k-n})$$

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

this is called the **characteristic equation** of the recurrence relation, and its solution is called **characteristic roots** of the recurrence relation.

- We will use characteristic roots to obtain **explicit formula** for the sequence.

A linear homogeneous recurrence relation of degree k with constant coefficients

- **Theorem 1.** Let c_1 and c_2 be real numbers.

Suppose that $r^2 - c_1 r - c_2 = 0$ has 2 distinct roots r_1 and r_2 . Then the sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}$$

if and only if

$$a_n = \alpha_1(r_1)^n + \alpha_2(r_2)^n \quad (1)$$

$n = 0, 1, \dots$, where α_1 and α_2 are constants.

Example 1

Fibonacci sequence is given by the following recurrence relation:

$$F_n = F_{n-1} + F_{n-2}, n \geq 2,$$

$$F_0 = 0, F_1 = 1.$$

Find the explicit formula for F_n .

Solution: Solve the characteristic equation:

$$r^2 - r - 1 = 0,$$

its characteristic roots are:

$$r_1 = \frac{1 + \sqrt{5}}{2}; \quad r_2 = \frac{1 - \sqrt{5}}{2}$$



Leonardo Fibonacci
1170-1250

Example 1

- The explicit formula:

$$F_n = \alpha_1 \cdot (r_1)^n + \alpha_2 \cdot (r_2)^n$$

where α_1, α_2 are constants and could be determined by using initial conditions F_0, F_1 .

$$F_0 = \alpha_1 + \alpha_2 = 0$$

$$F_1 = \alpha_1 r_1 + \alpha_2 r_2 = 1$$

Solving these two equations, we have: $\alpha_1 = \frac{1}{\sqrt{5}}; \quad \alpha_2 = -\frac{1}{\sqrt{5}}$

Muavre formula

Therefore

$$F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right), \quad n \geq 0.$$

The case: one characteristic root with multiplicities 2

But what happens if the characteristic equation has only one root?

Theorem 2: Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1 r - c_2 = 0$ has only one root r_0 . Then a sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$

if and only if

$$a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$$

$n = 0, 1, \dots$, where α_1, α_2 are constants.

Example 2

Solving the following recurrence relation

$$a_n = 6 a_{n-1} - 9 a_{n-2}$$

with initial conditions $a_0 = 1$ and $a_1 = 6$.

Solution:

Characteristic equation:

$r^2 - 6r + 9 = 0$ has one root $r = 3$. The solution is:

$$a_n = \alpha_1 3^n + \alpha_2 n 3^n$$

To determine α_1, α_2 , using the initial conditions, we have:

$$a_0 = 1 = \alpha_1,$$

$$a_1 = 6 = \alpha_1 * 3 + \alpha_2 * 1 * 3$$

Solving these equations, we have $\alpha_1 = 1$ and $\alpha_2 = 1$.

Therefore, the solution of the recurrence relation is:

$$\mathbf{a_n = 3^n + n 3^n}$$

General case

Theorem 3. Let $c_1, c_2, \dots, c_k$ be real numbers. Assume the characteristic equation

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_k = 0$$

has k distinct roots $r_1, r_2, \dots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k},$$

if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$$

where $n = 0, 1, 2, \dots$, and $\alpha_1, \alpha_2, \dots, \alpha_k$ are constants

Example 3

Solving the recurrence relation:

$$a_n = 6 a_{n-1} - 11 a_{n-2} + 6 a_{n-3}$$

where initial conditions

$$a_0 = 2, a_1 = 5, a_2 = 15.$$

Solution: Characteristic equation

$$r^3 - 6 r^2 + 11 r - 6 = 0$$

has 3 distinct roots $r_1 = 1, r_2 = 2, r_3 = 3$.

Therefore, the solution is

$$a_n = \alpha_1 1^n + \alpha_2 2^n + \alpha_3 3^n \text{ for some constants } \alpha_1, \alpha_2, \alpha_3$$

Example 3

Using the initial conditions, we have following equations to determine the value for constants $\alpha_1, \alpha_2, \alpha_3$:

$$a_0 = 2 = \alpha_1 + \alpha_2 + \alpha_3$$

$$a_1 = 5 = \alpha_1 + \alpha_2.2 + \alpha_3.3$$

$$a_2 = 15 = \alpha_1 + \alpha_2.4 + \alpha_3.9.$$

Solving above equations, we have

$$\alpha_1 = 1, \alpha_2 = -1 \text{ and } \alpha_3 = 2.$$

Therefore, the solution of the recurrence relation is

$$a_n = 1 - 2^n + 2 \cdot 3^n$$

General case

Given linear homogeneous recurrence relation of degree k with constant coefficients

Its characteristic equation is:

$$a_n = \sum_{i=1}^k c_i a_{n-i}$$

$$r^k - \sum_{i=1}^k c_i r^{k-i} = 0$$

Theorem 4: If the characteristic equation has t distinct roots $r_1, \dots, r_t$ with multiplicities $m_1, \dots, m_t$ ($m_1 + \dots + m_t = k$). Then:

$$a_n = (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1}) r_1^n + \\ (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1}) r_2^n + \dots \\ (\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1}) r_t^n$$

$$a_n = \sum_{i=1}^t \left(\sum_{j=0}^{m_i-1} \alpha_{i,j} n^j \right) r_i^n$$

where $n \geq 0$, and α_{ij} are constants.

Example 4

Solve the following recurrence relation:

$$c_n = -4c_{n-1} + 3c_{n-2} + 18c_{n-3}, \quad n \geq 3,$$

$$c_0 = 1; c_1 = 2; c_2 = 13.$$

Solution: Characteristic equation

$$r^3 + 4r^2 - 3r - 18 = (r - 2)(r + 3)^2 = 0$$

Hence, the solution of the recurrence relation:

$$c_n = \alpha_{10} 2^n + (\alpha_{20} + \alpha_{21} n)(-3)^n$$

where $\alpha_{10}, \alpha_{20}, \alpha_{21}$ are constants

Example 4

These constants are determined by using initial conditions:

$$0 = c_0 = \alpha_{10} 2^0 + (\alpha_{20} + \alpha_{21} \cdot 0) (-3)^0 = \alpha_{10} + \alpha_{20}$$

$$2 = c_1 = \alpha_{10} 2^1 + (\alpha_{20} + \alpha_{21} \cdot 1) (-3)^1 = 2\alpha_{10} - 3\alpha_{20} - 3\alpha_{21}$$

$$13 = c_2 = \alpha_{10} 2^2 + (\alpha_{20} + \alpha_{21} \cdot 2) (-3)^2 = 4\alpha_{10} + 9\alpha_{20} + 18\alpha_{21}$$

Solving three equations above, we have: $\alpha_{10} = 1$; $\alpha_{20} = -1$; $\alpha_{21} = 1$

The solution of recurrence relation is:

$$c_n = 2^n + (-1 + n)(-3)^n$$

Linear nonhomogeneous recurrence relation with constant coefficients

- Linear *nonhomogeneous* recurrence relation with constant coefficients) has the term $F(n)$ dependent on n (and independent on any value of a_i) :

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$$

Linear homogeneous recurrence relation

Nonhomogeneous term

Solving Linear nonhomogeneous recurrence relation

The following result is the basis for solving the nonhomogeneous recurrence relation:

- If $a_n = p(n)$ is a particular solution of linear nonhomogeneous recurrence relation

$$a_n = \left(\sum_{i=1}^k c_i a_{n-i} \right) + F(n)$$

Then the solution of linear nonhomogeneous recurrence relation is of the form:

$$a_n = p(n) + h(n),$$

where $a_n = h(n)$ is the solution of the linear homogeneous part:

$$a_n = \sum_{i=1}^k c_i a_{n-i}$$

Solving Linear nonhomogeneous recurrence relation

Then, we have the method for solving linear nonhomogeneous recurrence relation:

- 1) Find the solution $h(n)$ of linear homogeneous part.
- 2) Find particular solution $p(n)$ of linear nonhomogeneous recurrence relation.
- 3) The solution of linear nonhomogeneous recurrence relation is of the form:
 $a_n = h(n) + p(n)$
- 4) Determine the constants from equations obtained by initial conditions

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$$

Linear homogeneous recurrence relation

Nonhomogeneous term

Solving Linear nonhomogeneous recurrence relation

How to find particular solution $p(n)$ of linear nonhomogeneous recurrence relation?

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$$

Linear homogeneous recurrence relation

Nonhomogeneous term

There is no known general method for solving nonhomogeneous linear recurrence relation. However, we can develop a method for solving the special case:

$$F(n) = G(n) \times s^n = (b_t n^t + b_{t-1} n^{t-1} + \dots + b_0) s^n$$

where s is a constant; $G(n)$ is polynomial in n

- If s is not equal to characteristic root, then particular solution $p(n)$ is of the form $F(n)$
- If s is equal to characteristic root with multiplicities m , then particular solution $p(n)$ is of the form $n^m \times Q(n) \times s^n$ where $Q(n)$ is of the form $G(n)$

Example 1: Tower of Hanoi

Solving:

$$a_n = 2a_{n-1} + 1, n \geq 1;$$

$$a_1 = 1.$$

Solution:

(1) Linear homogeneous part: $a_n = 2a_{n-1}$

→ Characteristic equation: $r - 2 = 0$ has one root $r = 2$

→ Solution of linear homogeneous part is: $h(n) = c_1 2^n$

(2) Linear Nonhomogeneous part $F(n) = 1$, then the particular solution is of the form

$$p(n) = C$$

Substitute to the nonhomogeneous recurrent relation: $C = 2C + 1 \rightarrow C = -1$

Hence the particular solution is: $p(n) = -1$.

Example 1: Tower of Hanoi

Solving:

$$a_n = 2a_{n-1} + 1, n \geq 1;$$

$$a_1 = 1.$$

(3) The solution of linear nonhomogeneous recurrence relation:

$$a_n = h(n) + p(n) = c_1 2^n - 1$$

(4) Determine c_1 by using initial condition:

$$a_1 = c_1 2^1 - 1 = 1$$

$$\rightarrow c_1 = 1$$

Hence, the solution of linear nonhomogeneous recurrence relation:

$$a_n = 2^n - 1, n \geq 1.$$

Example 2

Solving recurrence relation:

$$a_n = a_{n-1} + n, n \geq 1; a_1 = 2$$

Solution:

(1) Linear homogeneous part: $a_n = a_{n-1}$

Characteristic equation $r - 1 = 0$ has the root $r = 1$

→ The solution of linear homogeneous part is: $h(n) = c_1 1^n$

(2) Nonhomogeneous part $F(n) = n \times 1^n$, and 1 is the root of multiplicities 1, thus the particular solution is of the form

$$p(n) = n^1(C_2 + C_3 n)1^n$$

Substitute to the recurrence relation, we have:

$$n(C_2 + C_3 n) = (n-1)[C_2 + C_3(n-1)] + n$$

Thus, $C_2 = 1/2$ and $C_3 = 1/2$. The particular solution is

$$p(n) = (n+1)n/2$$

Example 2

Solving recurrence relation:

$$a_n = a_{n-1} + n, n \geq 1; a_1 = 2$$

(3) The solution of linear nonhomogeneous recurrence relation is

$$a_n = c_1 + (n+1)n/2$$

(4) The constant c_1 is determined by using initial condition:

$$a_1 = c_1 + 1 = 2$$

$$\rightarrow c_1 = 1$$

The solution of linear nonhomogeneous recurrence relation is

$$a_n = 1 + (n+1)n/2, n \geq 1$$

Recurrence tree method

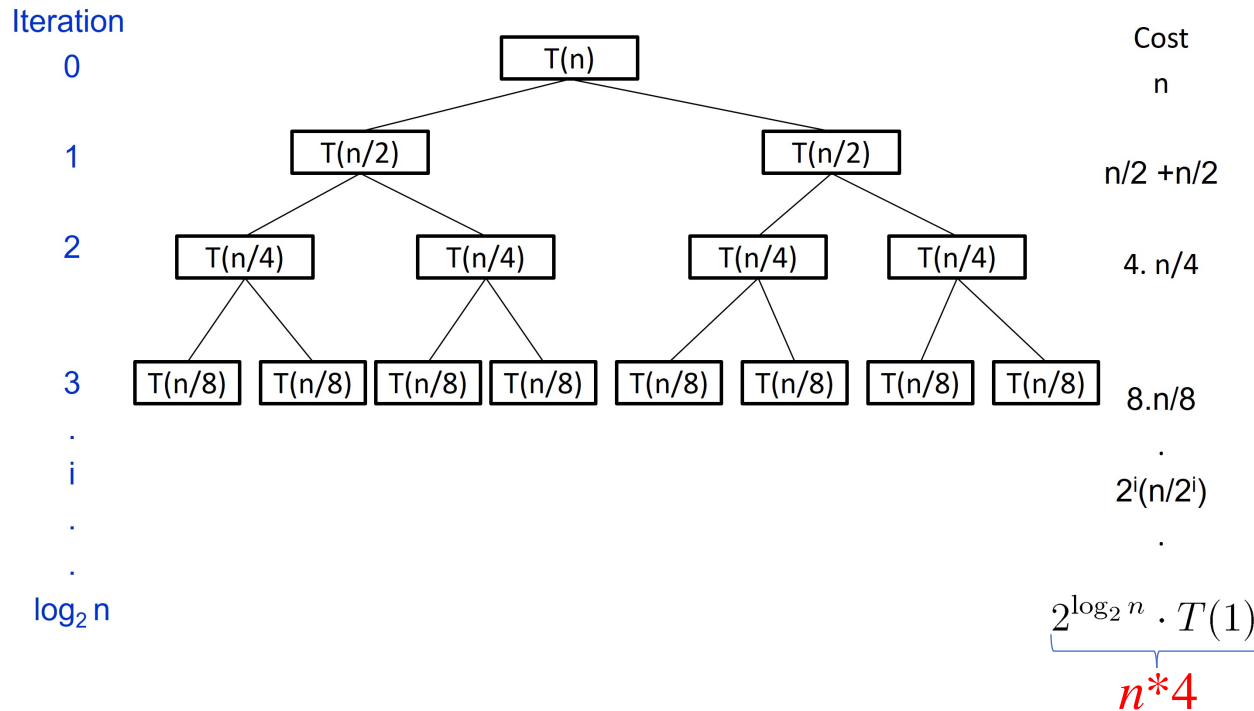
In this method, we draw a recurrence tree :

- Each node of the tree corresponds to an input data function. As going down the tree, the size of the input data will decrease.
- The last level of the tree corresponds to the smallest input data size.

We calculate the time taken by every level of tree, and finally, we sum the work done at all levels.

Example: Solving the recurrence relation:

$$T(n) = 2T(n/2) + n, \quad T(1) = 4$$



- The last level $\log_2 n$ has the value $= 4n$, thus we have:

$$T(n) = 4n + \sum_{i=0}^{(\log_2 n - 1)} 2^i \frac{n}{2^i} = n(\log n) + 4n$$

2.5. Recurrence relations and generating functions

2.5.1. Recurrence relations

2.5.2. Generating functions

2.5.2. Generating functions

Generating functions are a tool to solve a wide variety of counting problems and recurrence relations.

Example 1:

How many ways to give 12 oranges for three children: A, B and C such that: A gets at least four, and B and C gets at least two, but C gets no more than five

a, b, c are the number of oranges that A, B and C gets, respectively

Find the number of integer solutions to

$$a + b + c = 12 \text{ where } a \geq 4, b \geq 2, 2 \leq c \leq 5$$

2.5.2. Generating functions

Example 1:

Find the number of integer solutions to
 $a + b + c = 12$ where $a \geq 4, b \geq 2, 2 \leq c \leq 5$

Let

$$a: a(x) = x^4 + x^5 + x^6 + x^7 + x^8 \quad (\text{since } b + c \geq 4 \rightarrow a \leq 8)$$

$$b: b(x) = x^2 + x^3 + x^4 + x^5 + x^6 \quad (\text{since } a + c \geq 6 \rightarrow b \leq 6)$$

$$c: c(x) = x^2 + x^3 + x^4 + x^5 \quad (\text{since } 2 \leq c \leq 5)$$

The coefficient of x^{12} in $g(x) = a(x) \times b(x) \times c(x)$

$$= (x^4 + x^5 + x^6 + x^7 + x^8) (x^2 + x^3 + x^4 + x^5 + x^6) (x^2 + x^3 + x^4 + x^5)$$

is the solution to the problem

$g(x)$ is called a *generating function*.

Why???

$n=12$

=???

Assume x^a, x^b, x^c are terms derived from $(x^4 + x^5 + x^6 + x^7 + x^8)$, $(x^2 + x^3 + x^4 + x^5 + x^6)$, $(x^2 + x^3 + x^4 + x^5)$ respectively when developing the right-hand side $\rightarrow 4 \leq a \leq 8, 2 \leq b \leq 6, 2 \leq c \leq 5$

When developing the right-hand side, these three terms give us the term x^n where $n = a + b + c$:

$$g(x) = (x^4 + x^5 + x^6 + x^7 + x^8) (x^2 + x^3 + x^4 + x^5 + x^6) (x^2 + x^3 + x^4 + x^5)$$

$$= \dots + Kx^n + \dots$$

$\rightarrow$ The coefficient of x^n in $f(x)$ is the number of positive integer solutions to

$$a + b + c = n \text{ where } 4 \leq a \leq 8, 2 \leq b \leq 6, 2 \leq c \leq 5$$

2.5.2. Generating functions

Example 1:

Find the number of integer solutions to
 $a + b + c = 12$ where $a \geq 4, b \geq 2, 2 \leq c \leq 5$

Let

$$a: a(x) = x^4 + x^5 + x^6 + x^7 + x^8 \quad (\text{since } b + c \geq 4 \rightarrow a \leq 8)$$

$$b: b(x) = x^2 + x^3 + x^4 + x^5 + x^6 \quad (\text{since } a + c \geq 6 \rightarrow b \leq 6)$$

$$c: c(x) = x^2 + x^3 + x^4 + x^5 \quad (\text{since } 2 \leq c \leq 5)$$

The coefficient of x^{12} in $g(x) = a(x) \times b(x) \times c(x)$
 $= (x^4 + x^5 + x^6 + x^7 + x^8)(x^2 + x^3 + x^4 + x^5 + x^6)(x^2 + x^3 + x^4 + x^5)$

which is 14, is the solution

$g(x)$ is called a *generating function*.

1. $x^4 * x^3 * x^5$
2. $x^4 * x^4 * x^4$
3. $x^4 * x^5 * x^3$
4. $x^4 * x^6 * x^2$
5. $x^5 * x^2 * x^5$
6. $x^5 * x^3 * x^4$
7. $x^5 * x^4 * x^3$
8. $x^5 * x^5 * x^2$
9. $x^6 * x^2 * x^4$
10. $x^6 * x^3 * x^3$
11. $x^6 * x^4 * x^2$
12. $x^7 * x^2 * x^3$
13. $x^7 * x^3 * x^2$
14. $x^8 * x^2 * x^2$

2.5.2. Generating functions

Example 2: Four kinds of jelly beans: Red, Green, White, Black. In how many ways can we select 24 jelly beans so that we have an even number of white beans and at least six black ones?

White: $1 + x^2 + x^4 + \dots + x^{16} + x^{18}$

Black: $x^6 + x^7 + \dots + x^{16} + x^{18}$

Red (green): $1 + x^1 + x^2 + \dots + x^{16} + x^{18}$

$$g(x) = (1 + x^2 + x^4 + \dots + x^{16} + x^{18})(x^6 + x^7 + \dots + x^{16} + x^{18})(1 + x^1 + x^2 + \dots + x^{16} + x^{18})^2$$

The coefficient of x^{24} is the solution.

Example 3: How many nonnegative integer solutions are there for

$$c_1 + c_2 + c_3 + c_4 = 25?$$

$$g(x) = (1 + x^1 + x^2 + \dots + x^{24} + x^{25})^4$$

The coefficient of x^{25} is the solution

2.5.2. Generating functions

Definition: Let $a_0, a_1, a_2, \dots$ be a sequence of real numbers. The function

$$g(x) = a_0 + a_1x + \dots + a_kx^k + \dots = \sum_{i=0}^{\infty} a_i x^i$$

is called the *generating function* for the given sequence.

A finite sequence

$$a_0, a_1, a_2, \dots, a_n$$

can be regarded as the infinite sequence

$$a_0, a_1, a_2, \dots, a_n, 0, 0, \dots \text{ [set all terms higher than } n \text{ to } 0]$$

and its generating function

$$g(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

is a polynomial.

2.5.2. Generating functions

Example 4: For the binomial coefficients we already know that:

$$(x+y)^n = \sum_{k=0}^n C(n, k) x^k y^{n-k} \Rightarrow (x+1)^n = \sum_{k=0}^n C(n, k) x^k = g(x)$$

$(x+1)^n$ is the generating function for the sequence

$$C(n, 0), C(n, 1), \dots, C(n, k), \dots, C(n, n), 0, 0, 0 \dots$$

2.5.2. Generating functions

Definition: Let $a_0, a_1, a_2, \dots$ be a sequence of real numbers. The function

$$g(x) = a_0 + a_1x + \dots + a_kx^k + \dots = \sum_{i=0}^{\infty} a_i x^i$$

is called the *generating function* for the given sequence.

Some useful generating function:

Ex1: $(1 - x^{n+1}) = (1 - x)(1 + x + x^2 + x^3 + \dots + x^n)$.

So $(1 - x^{n+1})/(1 - x) = 1 + x + x^2 + x^3 + \dots + x^n$,

→ $(1 - x^{n+1})/(1 - x)$ is the generating function for the sequence:

1, 1, 1, ..., 1, 0, 0, 0, ... (where the first $n+1$ terms are 1)

Ex2: From Ex1, If $n \rightarrow \infty$ and $|x| < 1$, then $1 = (1 - x)(1 + x + x^2 + x^3 + \dots)$.

So, $1/(1 - x) = 1 + x + x^2 + x^3 + \dots$

→ $1/(1 - x)$ where $|x| < 1$ is the generating function for the sequence:

1, 1, 1, ..., 1, ...

2.5.2. Generating functions

If we have two generating functions $F(x)$ and $G(x)$, we define the sum and product as follows:

$$F(x) = \sum_{k=0}^{\infty} a_k x^k \quad G(x) = \sum_{k=0}^{\infty} b_k x^k$$

$$F(x) + G(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k \quad \leftarrow \text{Match all terms with equal powers in } x.$$

$$F(x)G(x) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k a_j b_{k-j} \right) x^k$$

Example 3: Example why multiplying generating functions is useful:

$$\begin{aligned} \frac{1}{(1-x)^2} &= \frac{1}{1-x} \frac{1}{1-x} = (1+x+x^2+x^3+\dots)(1+x+x^2+x^3+\dots) \quad \text{where } |x| < 1 \\ &= \left[\sum_{k=0}^{\infty} (1) x^k \right] * \left[\sum_{k=0}^{\infty} (1) x^k \right] = \sum_{k=0}^{\infty} \left(\sum_{j=0}^k 1 \right) x^k = \sum_{k=0}^{\infty} (k+1) x^k \end{aligned}$$

➔ $1/(1-x)^2$ is the generating function for the sequence 1,2,3,4,....

2.5.2. Generating functions

Generating functions are a tool to solve a wide variety of counting problems and **recurrence relations**.

Definition: Let $a_0, a_1, a_2, \dots$ be a sequence of real numbers. The function

$$g(x) = a_0 + a_1x + \dots + a_kx^k + \dots = \sum_{i=0}^{\infty} a_i x^i$$

is called the *generating function* for the given sequence.

2.5.2. Generating functions

Example: Solve the recurrence relation $a_n - 3a_{n-1} = n, n \geq 1, a_0 = 1$

$$g(x) - 1 - 3xg(x) = \frac{x}{(1-x)^2}$$

$$g(x) = \frac{1}{1-3x} + \frac{x}{(1-3x)(1-x)^2}$$

$$g(x) = \frac{1}{1-3x} + \frac{A}{(1-3x)} + \frac{B}{(1-x)} + \frac{C}{(1-x)^2}$$

$$\Rightarrow A = \frac{3}{4}; B = -\frac{1}{4}; C = -\frac{1}{2};$$

$$g(x) = a_0 + a_1x + \dots + a_kx^k + \dots = \sum_{i=0}^{\infty} a_i x^i$$

$$g(x) = \frac{7/4}{(1-3x)} + \frac{-1/4}{(1-x)} + \frac{-1/2}{(1-x)^2}$$

We find a_n by determining the coefficient of x^n in $g(x)$

➔ determining the coefficient of x^n in each of the three summands

2.5.2. Generating functions

Example: Solve the recurrence relation $a_n - 3a_{n-1} = n, n \geq 1, a_0 = 1$

$$g(x) = \frac{7/4}{(1-3x)} + \frac{-1/4}{(1-x)} + \frac{-1/2}{(1-x)^2}$$

We find a_n by determining the coefficient of x^n in each of the three summands

$$\begin{aligned} 1. \quad (7/4)/(1-3x) &= (7/4) [1/(1-3x)] \\ &= (7/4) [1 + 3x + (3x)^2 + (3x)^3 + \dots] \end{aligned}$$

→ The coefficient of x^n is $(7/4)3^n$

$$\begin{aligned} 2. \quad (-1/4)/(1-x) &= (-1/4) [1/(1-x)] \\ &= (-1/4) [1 + x + (x)^2 + (x)^3 + \dots] \end{aligned}$$

→ The coefficient of x^n is $(-1/4)$

$$\begin{aligned} 3. \quad (-1/2)/(1-x)^2 &= (-1/2) [1/(1-x)^2] \\ &= (-1/2) [1 + 2x + 3(x)^2 + 4(x)^3 + \dots] \end{aligned}$$

→ The coefficient of x^n is $(-1/2)(n+1)$

Therefore, $a_n = (7/4) 3^n - (1/2)n - 3/4, n \geq 0$